Tutorial Questions (Set-VI)

Q1 Let $f\colon [a,b] \to \mathbb{R}$ be a function, $|f(x)| \le M,\ \forall x \in [a,b]$. Let $P = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a,b]$. Suppose $I_i = [x_{i-1}, x_i]$ and $\Delta x_i = x_i - x_{i-1}$. For $1 \le i \le n$, set

$$m_i(f) = \inf_{t\in I_i} f(t) \quad \text{and} \quad M_i(f) = \sup_{t \in I_i} f(t).$$

Define functions $|f|$ and f^2 by $|f|(x) = |f(x)|$ and $f^2(x) = (f(x))^2,\ \forall x \in [a,b]$. Let $L(f,P) = \sum_{i=1}^n m_i(f)\Delta x_i$ and $U(f,P) = \sum_{i=1}^n M_i(f)\Delta x_i$ be the lower and upper sums of f w.r.t P. Let $\mathcal{R}[a,b]$ be the set of Riemann integrable functions on $[a,b]$.

(i) Show that $M_i(f) - m_i(f) = \sup\{|f(t) - f(s)| : t, s \in [x_{i-1}, x_i]\}$.

(ii) Show that $M_i(|f|) - m_i(|f|) \le M_i(f) - m_i(f)$.

(iii) Show that $M_i(f^2) - m_i(f^2) \le 2M(M_i(|f|) - m_i(|f|))$.

(iv) Show that $L(-f,P) = -U(f,P)$ and $U(-f,P) = -L(f,P)$. If P' is a refinement of P, then show that $L(f,P) \le L(f,P')$ and $U(f,P') \le U(f,P)$.

(v) Show that if $f \in \mathcal{R}[a,b]$, then $|f| \in \mathcal{R}[a,b]$ as well as $f^2 \in \mathcal{R}[a,b]$. However, $|f| \in \mathcal{R}[a,b]$ or $f^2 \in \mathcal{R}[a,b]$ need not imply $f \in \mathcal{R}[a,b]$.

(vi) If $f \in \mathcal{R}[a,b]$, then show that $\left| \int_a^b f\,dx \right| \ \le\ \int_a^b |f|\,dx \ \le\ M(b-a)$.

Q2 Let $f\colon [a,b] \to \mathbb{R}$ be a function. Prove the following statements.

(i) If f is continuous or monotonic, then $f \in \mathcal{R}[a,b]$.

(ii) The Dirichlet function on $[0,1]$ is not Riemann integrable, but the Thomae function on $[0,1]$ is Riemann integrable.

(iii) Let $f(x) = \begin{cases} 1 - \frac{1}{2^n} & \text{if } x \in [1 - \frac{1}{2^{n-1}}, 1 - \frac{1}{2^n}) \text{ for } n \ge 1, \\ 1 & \text{if } x = 1, \end{cases}$ be a function on $[0,1]$ called the *Zeno's staircase*. Then $f \in \mathcal{R}[0,1]$. Also, determine $\int_0^1 f\,dx$.

(iv) Define the *Cantor function* $h\colon [0,1] \to [0,1]$. Then h is monotonically increasing continuous function and hence $h \in \mathcal{R}[0,1]$. Also, determine $\int_0^1 h\,dx$.

Q3 (i) Let $f, g \in \mathcal{R}[a,b]$. Show that $f + g \in \mathcal{R}[a,b]$ and $fg \in \mathcal{R}[a,b]$.

(ii) Let $f\colon [a,b] \to \mathbb{R}$ be a function and $c \in (a,b)$. Show that if $f \in \mathcal{R}[a,b]$, then $f \in \mathcal{R}[a,c]$ and $f \in \mathcal{R}[c,b]$. Conversely, if $f \in \mathcal{R}[a,c]$ and $f \in \mathcal{R}[c,b]$, then show that $f \in \mathcal{R}[a,b]$. Further, in either of the cases,

$$\int_a^b f\,dx = \int_a^c f\,dx + \int_c^b f\,dx\ .$$

Q4 A subset $S \subseteq \mathbb{R}$ is a *zero set* (or a *measure zero set*) if for every $\epsilon > 0$, there is a countable family $\{(a_j, b_j) : j \in \Lambda\}$ (where $\Lambda \subseteq \mathbb{N}$) of open intervals such that $S \subseteq \bigcup_{j\in\Lambda}(a_j, b_j)$ and the total length of intervals $\sum_{j\in\Lambda} (b_j - a_j) < \epsilon$. Prove the following statements.

(i) A finite subset of $\mathbb{R}$ is a zero set. Also, a countable subset is a zero set.

(ii) Every subset of a zero set is a zero set.

(iii) If S_1 and S_2 are zero sets, then $S_1 \cup S_2$ is a zero set. Moreover, if $\{S_n : n \in \mathbb{N}\}$ is a countable family of zero sets, then the union $\bigcup_{n\in\mathbb{N}} S_n$ is a zero set.

(iv) The Cantor ternary set C is a zero set.

Q5 Let (X, d) be a metric space and $f\colon X \to \mathbb{R}$ be a function. For $x \in X$ and $r > 0$, we define $\Omega_f(x, B_d(x;r)) = \Omega_f(x, r) = \sup\{|f(u) - f(v)| : u, v \in B_d(x;r)\}$ and $\omega_f(x) = \inf\{\Omega_f(x, r) : r > 0\}$. We say that $\omega_f(x)$ is called the *oscillation of* f at x.

(i) Show that $\omega_f\colon X \to \mathbb{R}$ is upper semicontinuous i.e., $\omega_f^{-1}((-\infty, b))$ is open in X for every $b \in \mathbb{R}$.

(ii) Show that f is continuous at $x \in X$ if and only if $\omega_f(x) = 0$.

(iii) Show that the set of points of continuity of f is a G_δ-set. That is, $X \setminus E_f$ is a countable intersection of open sets. (E_f is the set of discontinuities of f.)

(iv) If $f \in \mathcal{R}[a, b]$, then by Riemann-Lebesgue theorem, show that f is continuous on a dense G_δ-set. (Hint : Use Baire category theorem)

Q6 (i) (Change of variable formula) Let $f \in \mathcal{R}[a, b]$. Suppose $\alpha\colon [c, d] \to [a, b]$ is a continuously differentiable, strictly increasing function. Then show that

$$\int_a^b f(x)\,dx = \int_c^d f(\alpha(u))\ \alpha'(u)\,du\ .$$

(ii) (Integration by parts) Let f, g be differentiable function on $[a, b]$ and derivatives $f', g' \in \mathcal{R}[a, b]$. Then show that

$$\int_a^b f(x)\ g'(x)\,dx = (f(b)g(b) - f(a)g(a)) - \int_a^b f'(x)\ g(x)\,dx\ .$$

(iii) Show that every continuous function on $[a, b]$ has a primitive.

(iv) Show that the unit-step function $J(x) = \begin{cases} 0 & \text{if } -1 \le x \le 0, \\ 1 & \text{if } 0 < x \le 1, \end{cases}$ has no primitive.

(v) The discontinuous function $f(x) = \begin{cases} 0 & \text{if } x \le 0, \\ \sin\left(\frac{\pi}{x}\right) & \text{if } x > 0, \end{cases}$ has a primitive on any interval $[a, b]$ with $a < 0$ and $b > 0$.